FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2015

PART TEST –III

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

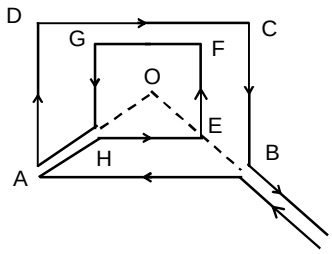
PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

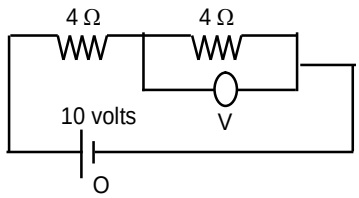
1. ABCD is a square of side 4R and EFGH is a square loop of side 2R. The loop carries a current I. The magnetic field at the centre O of the loop



(A) $\frac{\mu_0 I}{2\pi R}\odot$ (B) $\frac{\mu_0 I}{2\pi R}\otimes$

(C) $\frac{\mu_0 I}{\sqrt{2}\pi R}\otimes$ (D) $\frac{\mu_0 I}{\sqrt{2}\pi R}\odot$

2. In the given circuit, if the voltmeter is ideal its reading is V_1 volts. If the voltmeter has a resistance of 100Ω, the reading is V_2.

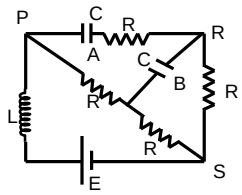


Then $|V_1 - V_2|$ is equal to :-

(A) Zero (B) 0.3volts.

(C) 0.2volts (D) 0.1volts

3. The charge on the capacitors A and B in steady state is :-



(A) $\frac{CE}{2}, CE$ (B) $\frac{CE}{2}, \frac{CE}{2}$

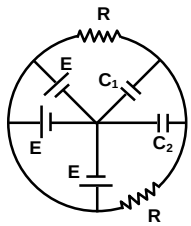
(C) $CE, \frac{CE}{2}$ (D) CE, CE

4. In a region, uniform electric field exists as $\vec{E} = \left[10\hat{i} + 10\hat{j}\right] N/C.$ If the potential of origin is 0 volts, the potential of point (10m, 10m, 10m) is

(A) 20 volts (B) – 200 volts

(C) 10 volts (D) – 10 volts

5. Find the current in resistor R at steady state

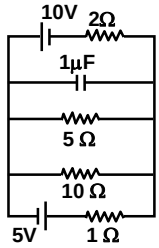


(A) $\frac{2E}{R}$ (B) zero

(C) $\frac{3E}{R}$ (D) $\frac{EC_1}{RC_2}$

Space for rough work

6. The charges on the plates of the capacitor in the steady state in the given circuit is



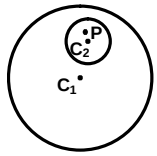
(A) zero. (B) 5 μC
(C) 10 μC (D) none of these

7. An infinite number of charges of magnitudes q, q/2, q/4, q/8 and so on are placed along the circumference of a circle of radius 1 unit with centre as origin. The potential at the origin will be

(A) $\frac{q}{2\pi\varepsilon_0}$ (B) $\frac{q}{4\pi\varepsilon_0}$

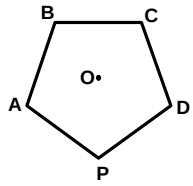
(C) $\frac{q}{3\pi\varepsilon_0}$ (D) infinite

8. From a uniform solid sphere an eccentric cavity has been cut out as shown in the figure. C_1 and C_2 represent their centres respectively. If the size of the cavity is increased keeping the centre at C_2, then the field strength at any internal point P will



(A) be zero
(B) remain constant in magnitude but direction of field will change with the change in position of P.
(C) remain constant in magnitude and direction for all positions of point P.
(D) gradually decrease in magnitude as the size of cavity increases.

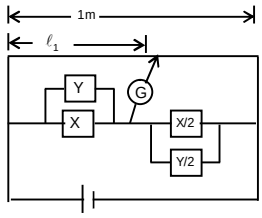
9. A regular pentagon having four identical charges of equal magnitude placed at the four corners A, B, C and D produce a field $\vec{E} = \left(-3\hat{i} + 4\hat{j} - 2\hat{k}\right)$ units at the centre O of the pentagon. If all the four charges are removed from the corners and reassembled at the fifth point P, find the field at O due to this rearrangement.



(A) $10\hat{i} - 5\hat{j} + 9\hat{k}$ (B) $3\hat{i} - 4\hat{j} + 2\hat{k}$

(C) $12\hat{i} - 16\hat{j} + 8\hat{k}$ (D) zero

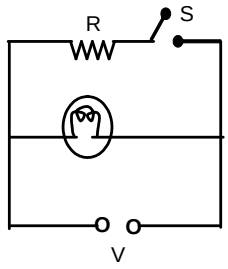
10. Figure shows a meter bridge. If there is no current through galvanometer then ℓ_1 is equal to



(A) 200/3 cm (B) 100/3 cm
(C) 50 cm (D) None of the above

Space for rough work

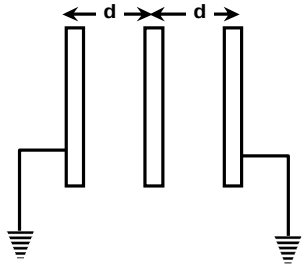
11. A bulb is connected across a voltage source. A resistor R is connected in parallel to the bulb through a switch S. If switch S is closed then
(A) Brightness of bulb decreases
(B) Brightness of bulb increases
(C) Brightness remains unchanged.
(D) None of the above



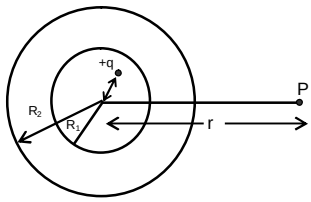
12. Three similar plates are arranged 'd' meters apart as shown in the figure. If area of each plate is A and a charge +Q is given to middle plate then potential of middle plate is

(A) $\frac{Qd}{A\in_0}$ (B) $\frac{Qd}{2A\in_0}$

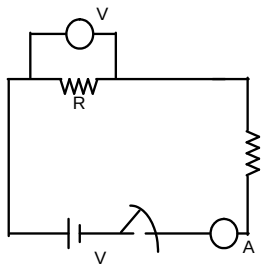
(C) $\frac{Qd}{4A\in_0}$ (D) $\frac{2Qd}{A\in_0}$



13. A conducting shell having inner radius R_1 and outer radius R_2 contains a charge + q which is placed at a distance x from its centre. Field at an exterior point P which is situated at a distance r from centre of shell $(r > R_2)$
(A) depends on x
(B) depends on R_1 and R_2
(C) depends only or r
(D) None.

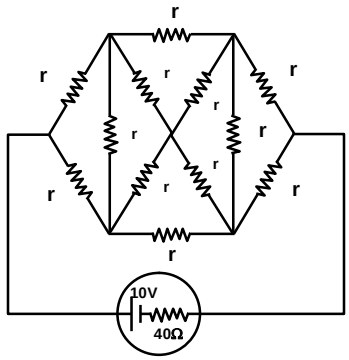


14. The potential deference across the resistor as measured by a voltmetre, is 60 ± 1.5V and the current through the resistor as measured by an ammeter is (2 ± 0.01) A. what is the maximum% error in the measured value of R?
(A) 3% (B) 1.5%
(C) 2.5% (D) None of these



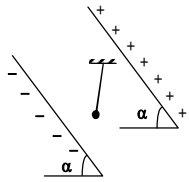
Space for rough work

15. Find the value of r so that maximum power is generated in external circuit.



(A) r = 10 Ω (B) r = 20 Ω
(C) r = 30 Ω (D) r = 40 Ω

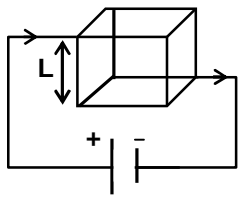
16. Consider a parallel plate capacitor having an electric field E inside it as shown in the adjacent figure. A particle of mass m and charge q is hanging inside the capacitor through a light inextensible string of length ℓ. The time period of this pendulum is



(A) $T = 2\pi\sqrt{\frac{\ell}{g+(qE/m)}}$ (B) $T = 2\pi\sqrt{\frac{\ell}{g^2+\frac{q^2E^2}{m^2}-\frac{2qgE}{m}}}$

(C) $T = 2\pi\sqrt{\frac{\ell}{g^2+\frac{q^2E^2}{m^2}+\frac{2qgE}{m}}}$ (D) none of the above

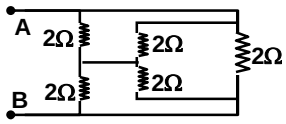
17. A cube made from wires of equal length is connected to a battery as shown in the figure. The magnetic field at the centre of the cube is



(A) $\frac{12\mu_0 I}{\sqrt{2}\,\pi L}$ (B) $\frac{6\mu_0 I}{\sqrt{2}\,\pi L}$

(C) $\frac{6\mu_0 I}{\pi L}$ (D) zero

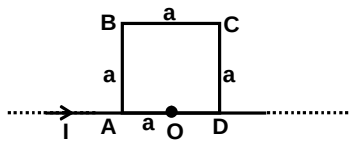
18. Find the equivalent resistance across AB:



(A) 1 Ω (B) 2 Ω
(C) 3 Ω (D) 4 Ω

Space for rough work

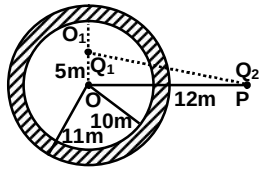
19. The resistance per unit length of the wires in segments AB, BC, CD, and AD are the same. The magnetic field due to the current at point O is



(A) $\frac{3\mu_0 I}{4\sqrt{5}\pi a}$ (B) $\frac{3\mu_0 I}{\sqrt{5}\pi a}$

(C) $\frac{\mu_0 I}{2\sqrt{5}\pi a}$ (D) none of the above

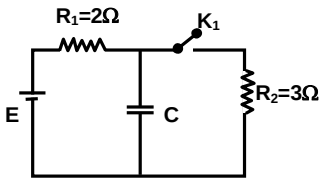
20. A charge Q_1 is placed at O, inside a hollow conducting sphere having inner and outer radii as 10 m and 11 m as shown. The force experienced by Q_2 at P is F_1 and force experienced by Q_2 when Q_1 is placed at O_1 is F_2. Then F_1/F_2 is equal to



(A) 1 (B) $\left(\frac{12}{13}\right)^2$

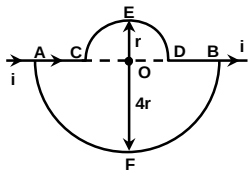
(C) $\left(\frac{13}{12}\right)^2$ (D) none of the above

21. In the circuit shown in the figure k_1 is open. The charge on capacitor C in steady state is q_1. Now key is closed and at steady state charge on C is q_2. The ratio of charges q_1/q_2 is



(A) 5/3 (B) 3/5
(C) 1 (D) 2/3

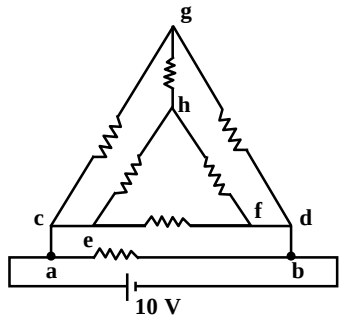
22. In the shown in the figure AC and BD are straight lines and CED and AFB are semicircular with radii r and 4r respectively. The entire setup is lying in the same plane. If i is current entering at A what fraction of i will flow in the ACEDB such that resultant magnetic field at O is zero.



(A) i/5 (B) 4i/5
(C) 3i/5 (D) no current through bigger semicircular part.

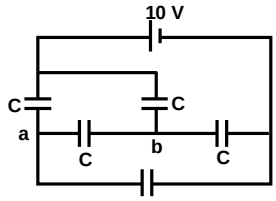
Space for rough work

23. Consider the network shown in the figure. All resistances are equal to 2Ω. What is the potential difference between points e and h?



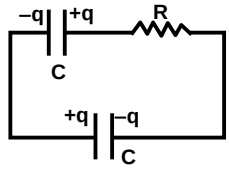
(A) zero (B) 5V
(C) 2.5 V (D) 10 V

24. What is the energy stored in the capacitor between terminals a and b of the network shown in the figure? (Capacitance of each capacitor C = 5μF).



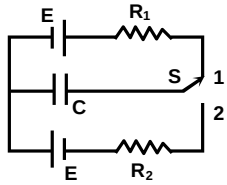
(A) 1 μJ (B) 0.25 μJ
(C) zero. (D) 15.6 μJ

25. Two capacitors having equal capacitance C, are having same charges q each. They are connected across a resistor R as shown in the figure. What is the total heat generated in the resistor upto the instant when the steady state has reached?



(A) $q^2/2C$ (B) q^2/C
(C) zero (D) none of the above.

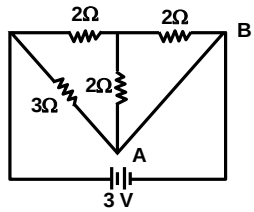
26. In the circuit shown, switch is placed in position1, till the capacitor is charged to half of the maximum possible charge in this situation. Now the switch S is placed in position 2. The maximum energy lost by the circuit after switch S is placed in position 2 is



(A) $\frac{1}{2}CE^2$ (B) $\frac{1}{8}CE^2$
(C) $\frac{7}{8}CE^2$ (D) $\frac{9}{8}CE^2$

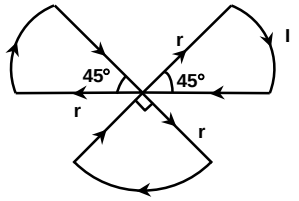
Space for rough work

27. The current in branch AB is



(A) infinite (B) 1.33A
(C) 2 A (D) 1.5 A

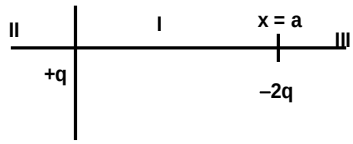
28. Calculate the magnetic field at point P due to three arcs as shown in the figure



(A) $\frac{\mu_0 I}{2r}$ (B) $\frac{\mu_0 I}{4r}$

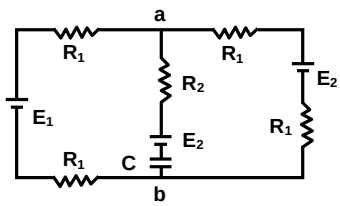
(C) $\frac{\mu_0 I}{2\pi r}$ (D) zero

29. A point charge +q is located at the origin and a point charge – 2q is located at x = a (a > 0). In which of the following region or regions might there exists a point where the electric potential (V) is zero?



(A) $-\infty < x < a$ (B) $0 < x < a$
(C) $a < x < \infty$ (D) V does not vanish in the region $-\infty < x < \infty$

30. What is the potential difference between points a and b, in the steady state in the circuit given below? (E_1 = 2.1V, E_2 = 6.3 V, R_1 = 1.7 Ω, R_2 = 3.5 Ω, C = 2μF)



(A) 4.9 V (B) 2.8 V
(C) 5.9 V (D) 4.2 V

Space for rough work

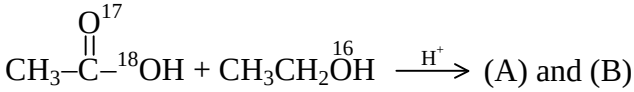
Chemistry PART – II

SECTION – A

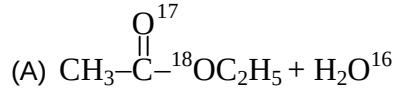
Straight Objective Type

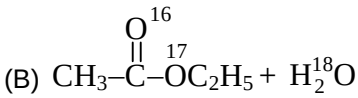
This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

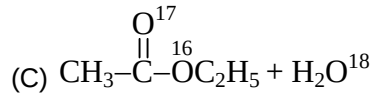
1. Consider the reaction,



(A) and (B) respectively are

(A) 

(B) 

(C) 

(D) Both (A) and (B)

2. The product of the given reaction $CH_3COOAg \xrightarrow{I_2/\Delta}$ [X]

[X] is

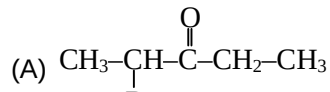
(A) $CH_3–I$

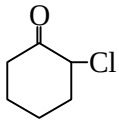
(B) CH_3COOCH_3

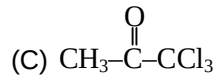
(C) $CH_3–CH_3$

(D) $CH_3–COOC_2H_5$

3. Which halo derivative(s) on treatment with $\overset{\ominus}{O}H$ will give carboxylic acid?

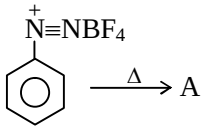
(A) 

(B) 

(C) 

(D) All of these

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4. 

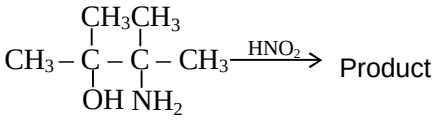
In the above process product A is

(A) Fluorobenzene
(B) Benzene
(C) 1,4-difluorobenzene
(D) 1,3-difluorobenzene

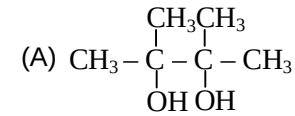
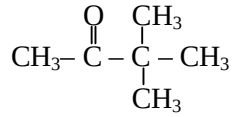
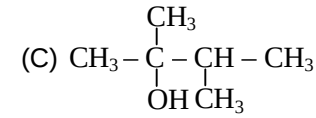
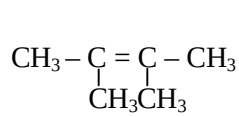
5. Treatment of NH_3 with excess of ethyl chloride gives:

(A) diethylamine
(B) ethane
(C) tetraethylammonium chloride
(D) methylamine

6. The product of the given reaction,



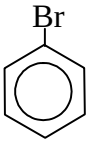
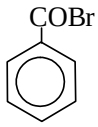
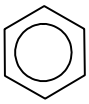
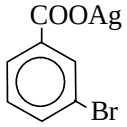
is

(A) 
(B) 
(C) 
(D) 

7. Which among the following amines will give carbylamine reaction?

(A) $CH_3–CH_2–NH_2$
(B) $CH_3–NH–CH_3$
(C) $(C_6H_5)_3N$
(D) $CH_3–CH_2–NH–OH$

8. Silver benzoate reacts with bromine in acetone to form

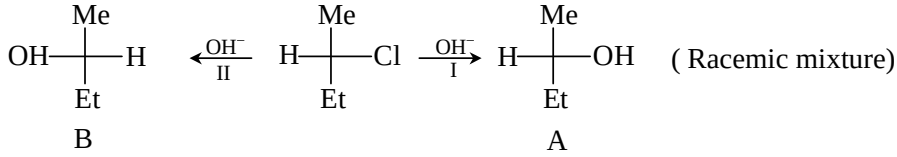
(A) 
(B) 
(C) 
(D) 

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9. Which of the following compound is most rapidly hydrolyzed by S_N 1 mechanism?

(A) C_6H_5Cl (B) $Cl-CH_2-CH=CH_2$

(C) $(C_6H_5)_3CCl$ (D) $C_6H_5CH_2Cl$

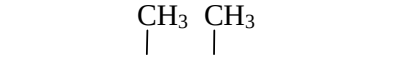
10. 

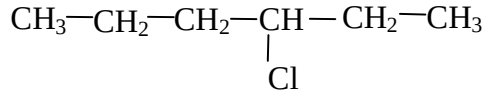
Steps I and II are:

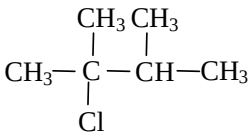
(A) both S_N1 (B) both S_N2

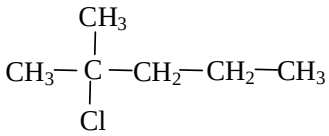
(C) I S_N1, II S_N2 (D) I S_N2, II S_N1

11. An alkyl halide (X) having molecular formula $C_6H_{13}Cl$ reacts with $(CH_3)_3C\overset{\ominus}{O}\overset{\oplus}{K}$ to form two isomeric alkenes (Y) and (Z) of molecular formula (C_6H_{12}). Both the alkenes give 2,3–Dimethyl butane on hydrogenation. The halide (X) is

(A) 

(B) 

(C) 

(D) 

12. The formation of cyanohydrin from a ketone is an example of:

(A) nucleophilic substitution. (B) electrophilic substitution.

(C) electrophilic addition. (D) nucleophilic addition.

13. Acetone is mixed with bleaching powder to give:

(A) chloroform (B) acetaldehyde

(C) ethanol (D) phosgene

14. Which of the following compounds is stable?

(A) $CH_3CH(OH)_2$ (B) $(CH_3)_2CH(OH)_2$

(C) $CCl_3CH(OH)_2$ (D) none of these

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15. Which of the following does not respond to haloform reaction?

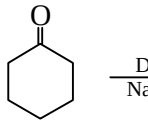
(A) CH_3-CH_2-OH

(B) CCl_3CHO

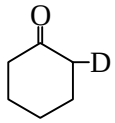
(C) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-COC_2H_5$

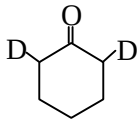
(D) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

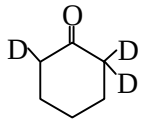
16. In the given reaction

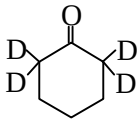


$\xrightarrow[NaOD]{D_2O}$ (P) will be

(A) 

(B) 

(C) 

(D) 

17. A compound (X) C_4H_8O which gives a 2, 4-DNP derivative and a positive iodoform test is:

(A) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

(B) $CH_3-CH_2-CH_2-CHO$

(C) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CHO$

(D) All of these

18. The repeat unit of polystyrene is

(A) $-CH_2-\underset{\underset{C_6H_5}{|}}{CH}-$

(B) $-CH_2-\underset{\underset{CN}{|}}{CH}-$

(C) $-CH_2-\underset{\underset{Cl}{|}}{CH}-$

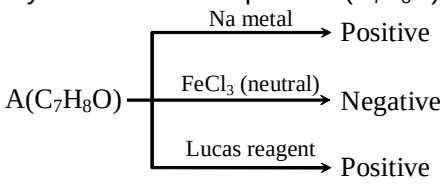
(D) $-CCl_2-CCl_2-$

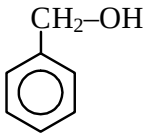
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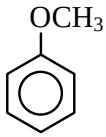
19. Which one of the following polymers may be classified as step growth polymer?
(A) Teflon (B) Polythene
(C) PVC (D) Nylon 66

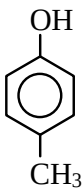
20. Propyne and propene can be distinguished by
(A) conc. H_2SO_4 (B) Br_2 inCCl_4
(C) $AgNO_3$ [Ammonical] (D) dil. $KMnO_4$

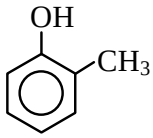
21. Identify an aromatic compound A(C_7H_8O) which gives following tests with the given reagents.



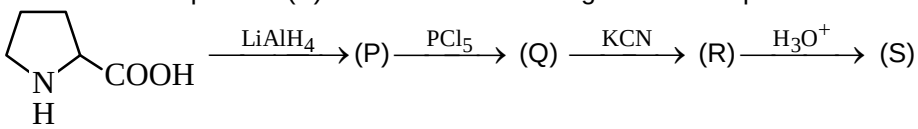
(A) 

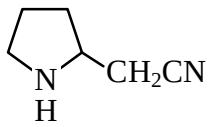
(B) 

(C) 

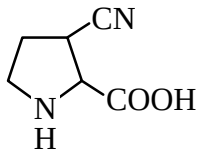
(D) 

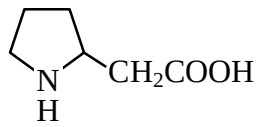
22. Predict the final product (S) formed in the following reaction sequence.



(A) 

(B) $H_2N(CH_2)_3CH_2CH_2COOH$

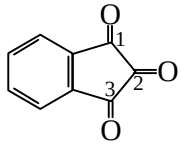
(C) 

(D) 

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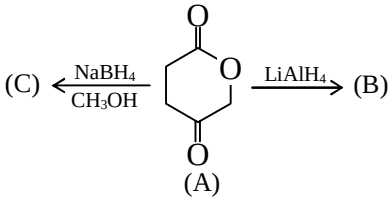
23. Carboxylic acids react with diazomethane to yield:
(A) Amines (B) Alcohols
(C) Esters (D) Amides

24. Which carbonyl group of the given compound is most reactive for nucleophilic addition reaction?

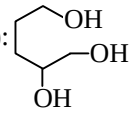
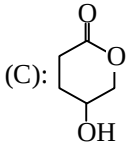


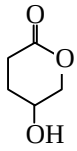
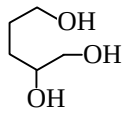
(A) 1 (B) 2
(C) 3 (D) All have equal reactivity

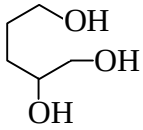
25. Identify the products (B) and (C) formed in the following set of reactions.

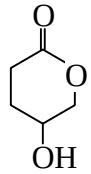


(A)

(A) (B):  (C): 

(B) (B):  (C): 

(C) Both (B) and (C): 

(D) Both (B) and (C): 

26. Identify the unknown products (A) and (B) in the given reaction:

Glucose + 3 $C_6H_5NH–NH_2$ $\xrightarrow{H^{\oplus}/\Delta}$ Osazone + (A) + (B)

(A) $C_6H_5NH_2$ and NH_3 (B) $C_6H_5NH_2$ and NH_2OH
(C) C_6H_5NH—NHOH and NH_3 (D) NH_2OH and H_2O

Space for rough work

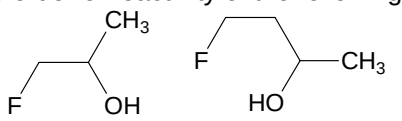
27. Buna–S rubber is a polymer of
(A) 1, 3-butadiene and styrene
(B) vinyl acetate
(C) Acrylonitrile
(D) none of these

28. Identify the final product (Z) in the following sequence of reactions:

$$CH_3NH_2 \xrightarrow[\text{excess}]{CH_3I} X \xrightarrow{AgOH} Y \xrightarrow{\Delta} Z + CH_3OH$$

(A) $CH_3 - CH_2 - CH_2 - NH_2$
(B) $(CH_3)_3N$
(C) $(CH_3)_2NH$
(D) $(CH_3)_4N^+OH^-$

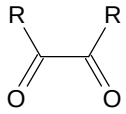
29. The order of reactivity of the following alcohols towards conc. HCl is

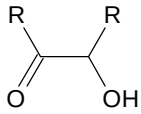


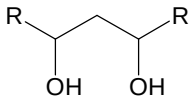
I II III IV

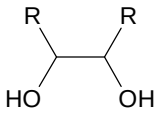
(A) I > II > III > IV
(B) I > III > II > IV
(C) IV > III > II > I
(D) IV > III > I > II

30. Which of the following will be oxidized by HIO_4?

(1) 

(2) 

(3) 

(4) 

Select the correct answer using the codes below:
(A) 1, 2 and 3
(B) 1, 3 and 4
(C) 1, 2 and 4
(D) 2, 3 and 4

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x}dx = \frac{f(x)}{(\sin x)^7} + c$, then f(x) is equal to

   (A) sin x (B) cos x
   (C) tan x (D) cot x

2. $\int \tan^{-1}\alpha \tan x \, d\alpha$ is equal to

   (A) $\tan^{-1}\alpha \sec x + c$ (B) $\tan^{-1}\alpha \ \ell n(\sec x) + c$
   (C) $\tan x\left\{\alpha\tan^{-1}\alpha - \frac{1}{2}\ell n\left(1+\alpha^2\right)\right\} + c$ (D) $\tan\alpha\left\{x\tan^{-1}x - \frac{1}{2}\ell n\left(1+x^2\right)\right\} + c$

3. If $k = \int_0^1 \frac{e^t}{1+t}dt$, then $\int_0^1 e^t \ell n(1+t)dt$ is equal to

   (A) k (B) 2k
   (C) $e\ell n2 - k$ (D) none of these

4. If $I_n = \int e^x (\sin x)^n dx$, then $\frac{I_3}{I_1}$ is equal to

   (A) $\frac{3}{5}$ (B) $\frac{1}{5}$
   (C) 1 (D) $\frac{2}{5}$

Space for rough work

5. If $k \in N$ and $I_k = \int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]\,dx$, (where [.] denotes the greatest integer function) then $\sum_{k=1}^{100} I_k$ equal to
(A) –10100 (B) –40400
(C) –20200 (D) none of these

6. Consider a function 'f' defined on the set of all non-negative integers such that f(0) = 1, f(1) = 0 and f(n) + f(n – 1) = nf(n – 1) + (n – 1)f(n – 2) for $n \geq 2$, then f(5) is equal to
(A) 40 (B) 44
(C) 45 (D) 60

7. Let R be the set of all real numbers the function f: $R \to R$ defined by $f(x) = x^3 - 3x^2 + 6x - 5$ is
(A) one-to-one but not onto (B) onto but not one-to-one
(C) neither one-to-one nor onto (D) one-to-one and onto

8. A function f: $R \to R$ is differentiable and satisfies the $f\left(\frac{1}{n}\right) = 0$ for all integers $n \geq 1$, then
(A) f(x) = 0 for all $n \in (0, 1]$ (B) f(0) = 0 = f ′(0)
(C) f(0) = 0 but f ′(0) need not be equal to (D) $|f(x)| \leq 1$ for all $x \in [0, 1]$

9. The maximum value of the sum of the intercepts made by any tangent to the curve ($a \sin^2 \theta$, $2a \sin \theta$) with the axes is
(A) 2a (B) $\frac{a}{4}$
(C) $\frac{a}{2}$ (D) a

10. $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$ where $0 < x \leq 1$, then in this interval
(A) f(x) and g(x) both are increasing (B) f(x) is decreasing and g(x) is increasing
(C) f(x) is increasing and g(x) is decreasing (D) none of these

11. $\int_{-1}^{1} \frac{\log\left(x+\sqrt{1+x^2}\right)}{x+\log\left(x+\sqrt{1+x^2}\right)} f(x)\,dx - \int_{-1}^{1} \frac{\log\left(x+\sqrt{1+x^2}\right)}{x+\log\left(x+\sqrt{1+x^2}\right)} f(-x)\,dx$ is equal to
(A) 0 (B) $2\int_{0}^{1} \frac{\log\left(x+\sqrt{1+x^2}\right)}{x+\log\left(x+\sqrt{1+x^2}\right)} \{f(x) - f(-x)\}\,dx$
(C) 2f(x) (D) none of these

Space for rough work

12. $\int e^x \left(\tan^{-1} x + \frac{2x}{\left(1+x^2\right)^2} \right) dx$ is equal to

(A) $e^x \left(\tan^{-1} x - \frac{1}{1+x^2} \right) + c$ (B) $e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) + c$

(C) $e^x \left(\cot^{-1} x - \frac{1}{1+x^2} \right) + c$ (D) $e^x \left(\tan^{-1} x + \frac{2}{1+x^2} \right) + c$

13. $\int \left(x^2 + x\right)\left(x^{-8} + 2x^{-9}\right)^{1/10} dx$ is

(A) $\frac{5}{11}\left(x^2 + 2x\right)^{11/10} + c$ (B) $\frac{5}{11}\left(x+1\right)^{11/10} + c$

(C) $\frac{6}{7}\left(x+1\right)^{11/10} + c$ (D) $\frac{11}{5}\left(x^2 + 2x\right)^{10/10} + c$

14. If $\int_0^1 \alpha e^{\beta x^2} \sin\left(x+k\right) dx = 0$ (for some $\alpha, \beta \in R$), $\alpha \neq 0$, then the value of k can belong to

(A) $\left[\frac{\pi}{3}, \frac{5\pi}{12}\right]$ (B) $\left[\frac{\pi}{3}, \frac{\pi}{2}\right]$

(C) $\left[\frac{3\pi}{4}, \frac{5\pi}{6}\right]$ (D) $\left[-\frac{\pi}{2}, -\frac{\pi}{3}\right]$

15. $\int_0^{[x]/3} \frac{8^x}{2^{[3x]}} dx$ (where [.] denotes the greatest integer function) is equal to

(A) $\frac{[x]}{\ell n\, 2}$ (B) $\frac{[x]}{\ell n\, 4}$

(C) $\frac{2[x]}{\ell n\, 2}$ (D) $\frac{[x]}{\ell n\, 8}$

16. Domain of $\left(\sqrt{x^2 - 4x + 3} + 1\right)\log_5\left(\frac{x}{5}\right) + \frac{1}{x}\left(\sqrt{8x - 2x^2 - 6} + 1\right) \leq 0$ are

(A) $(-\infty, 1] \cup [3, \infty)$ (B) [1, 3]

(C) {1, 3} (D) {1}

Space for rough work

17. Let f: R $\to$ R and g: R $\to$ R be two one-one and onto function such that they are the mirror images of each other about the line y = a. If h(x) = f(x) + g(x), then h(x) is
(A) bijective function
(B) injective but n of surjective
(C) surjective but not injective
(D) neither injective nor surjective

18. The set of value of α so that $f(x) = \frac{x-1}{\alpha - x^2 + 1}$ does not take any value in the interval $\left[-1, -\frac{1}{3}\right]$ is
(A) $\left(-\infty, -\frac{1}{4}\right]$
(B) $[2, \infty)$
(C) $\left(-\frac{1}{4}, 2\right)$
(D) $\left(-\infty, -\frac{1}{4}\right] \cup [2, \infty)$

19. Period of the function f(x) = cos $2\pi\{2x\}$ + sin $\pi\{2x\}$ is ((where {.} denotes the fractional part of function)
(A) 1
(B) $\frac{\pi}{2}$
(C) 1/2
(D) π

20. If $f(n+2) = \frac{1}{2}\left\{f(n+1) + \frac{9}{f(n)}\right\}$, n $\in$ N and f(x) > 0 for all n $\in$ N then $\lim_{n\to\infty} f(n)$ is equal to
(A) 3
(B) –3
(C) $\frac{1}{2}$
(D) none of these

21. The value of $\lim_{x\to 0}\left[\frac{x^2}{\sin x \tan x}\right]$ (where [.] denotes the greatest integer function) is
(A) 0
(B) 1
(C) limit does not exist
(D) none of these

22. The left hand derivative of f(x) = [x] sin (πx), at x = k, where k is an integer is (where [.] denotes the greatest integer function)
(A) $(-1)^k (k-1)\pi$
(B) $(-1)^{k-1} (k-1)\pi$
(C) $(-1)^k k\pi$
(D) $(-1)^{k-1} k\pi$

Space for rough work

23. Let $f(x)=\begin{cases} x^2\sin\left(\frac{\pi}{x}\right), & -1<x<1\ (x\neq 0) \\ x|x|, & x>1 \text{ or } x<-1 \end{cases}$ then

(A) f(x) is an odd function
(B) f(x) is an even function
(C) f(x) is neither odd nor even
(D) none of these

24. Given the function f(x) such that $2f(x)+xf\left(\frac{1}{x}\right)-2f\left(\left|\sqrt{2}\sin\left\{\pi\left(x+\frac{1}{4}\right)\right\}\right|\right)=4\cos^2\frac{\pi x}{2}+x\cos\frac{\pi}{x}$

(A) $f(2)+f\left(\frac{1}{2}\right)=0$
(B) f(1) = –1, but f(2), $f\left(\frac{1}{2}\right)$ can't be determined
(C) $f(2)+f(1)=f\left(\frac{1}{2}\right)$
(D) f(2) + f(1) = 1

25. Let $f(x)=\frac{1}{x}$ and $g(x)=\frac{1}{\sqrt{x}}$ then

(A) f{g(x)} and g{f(x)} have different domain
(B) f{g(x)} and g{f(x)} have same domain
(C) g{f(x)} is a bijective mapping
(D) f{g(x)} is neither odd nor even

26. The period of the function f(x) = sin 3x cos [3x] – cos 3x sin [3x] (where [.] denotes the greatest integer function) is

(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) 1
(D) 3

27. The integer n for which $\lim\limits_{x\to 0}\left(\frac{(\cos x-1)\left(\cos x-e^x\right)}{x^n}\right)$ is a finite non-zero number is

(A) 1
(B) 2
(C) 3
(D) 4

Space for rough work

28. If $f(x) = x^3 - x^2 + 100x + 2015$, then

(A) $f\left(\frac{1}{2015}\right) > f\left(\frac{1}{2016}\right)$ (B) $f(2015) > f(2016)$

(C) $f(2016) = f(2015)$ (D) $f\left(\frac{1}{2015}\right) = f\left(\frac{1}{2016}\right)$

29. The lines tangent to the curve $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ. Then θ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

30. If the function $f(x) = ax^3 + bx^2 + 11x - 6$ satisfies conditions of Rolle's theorem in [1, 3] for $x = 2 + \frac{1}{\sqrt{3}}$, then value of a and b are respectively

(A) –3, 2 (B) 2, –4

(C) 1, –6 (D) none of these

Space for rough work